



CMSC335: Web Application Development with JavaScript

Term: Summer 2026

Lecturer: Elias Gonzalez

Pronouns: He/Him

Email: egonzal5@umd.edu

Office Hours:

- TBA

Credits: 3

Course Dates: From Jun 1st, 2026 - July 24th, 2026

Course Times:

Lecture: Asynchronous, Lectures posted on ELMS

Classroom: ONLINE

Course Description

Web Application Development with JavaScript provides an introduction to modern ways of developing Web Applications/Services using JavaScript for both front-end and back-end. The course covers topics on fundamental JavaScript language constructs, server-side JavaScript, back-end data persistence, and client-side JavaScript to build Web Applications that interact with Web services and back-end databases. The course assumes you have never done HTML, CSS, JavaScript, and Node.js.

By the end of the course, you will be able to develop and host an online application that allows users to enter/retrieve/process data using a server. The server could access data from a database (MongoDB). For example, you can have a summer camp application where users can apply to a summer camp, check their submission status, etc. using a form. System administrators will be able to search current applications.

Learning Outcomes

After successfully completing this course you will be able to:

- Understand and develop web pages using HTML and CSS features.
- Use JavaScript language constructs for developing dynamic web pages.
- Understand the behavior (interaction with web pages) of code written in JavaScript.
- Use JavaScript to develop web services that interacts with 3rd-party services via web APIs.
- Understand the mechanism to provide data persistence in the back end (e.g., MongoDB).
- Use client-side JavaScript framework/libraries to develop single page applications (SPAs) interacting with web APIs.
- Be able to create webpages with React.

Required Resources

- Course Website: elms.umd.edu
- Application/Software:
 - VSCode
 - Google Chrome

Course Structure

This course will primarily be online, with recordings posted in batches (1-2 weeks at a time). Students are expected to watch the lectures by the end of the week they are for, in anticipation of projects and assessments. The course will consist of a few assessments: Coding Projects, “Semester” Projects, short quizzes and a Mid-Point Oral Exam.

Policies and Resources for Undergraduate Courses *(delete if you are teaching a graduate course)*

It is our shared responsibility to know and abide by the University of Maryland’s policies that relate to all courses, which include topics like:

- Academic integrity
- Student and instructor conduct
- Accessibility and accommodations
- Attendance and excused absences
- Grades and appeals
- Copyright and intellectual property

Please visit www.ugst.umd.edu/courserelatedpolicies.html for the Office of Undergraduate Studies’ full list of campus-wide policies and follow up with me if you have questions.

Course Guidelines

Names/Pronouns and Self-Identifications:

The University of Maryland recognizes the importance of a diverse student body, and we are committed to fostering inclusive and equitable classroom environments. I invite you, if you wish, to tell us how you want to be referred to in this class, both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). Keep in mind that the pronouns someone uses are not necessarily indicative of their gender identity. Visit trans.umd.edu to learn more.

Additionally, it is your choice whether to disclose how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. I will do my best to address and refer to all students accordingly, and I ask you to do the same for all of your fellow Terps.

Communication with Instructor:

Email: If you need to reach out and communicate with me, please email me at egonzal5@umd.edu. Please DO NOT email me with questions that are easily found in the syllabus or on ELMS (i.e. When is this assignment due? How much is it worth? etc.) but please DO reach out about personal, academic, and intellectual concerns/questions. While I will do my best to respond to emails within 48 hours.

Additionally, if you are requesting a logistical adjustment (extension, makeup, illness etc.) please email egonzal5@umd.edu with your concern.

ELMS: I will send IMPORTANT announcements via PIAZZA not ELMS messaging. Grades will be published on ELMS. You must make sure that your email & announcement notifications (including changes in assignments and/or due

dates) are enabled in ELMS so you do not miss any messages. You are responsible for checking your email and Piazza with regular frequency.

	Ask In/After Class	Go to Office Hours and ask a TA/Elias	Public Piazza Post	Private Piazza Post to all instructors	Email Elias	Message on ELMS
Quick Content or Clarification Questions (for the benefit of the class and yourself)	✓	✓	✓	✓	✗	✗
Specific project or quiz questions that reveal your approach and/or code	✗	✓	✗	✓	✓ *Office hours is preferable	✗
Extension Requests/Personal Emergencies and Requests	✗	✗	✗	✗	✓	✗
Grade Questions	✗	✗	✗	✗	✓	✗

Communication with Peers:

With a diversity of perspectives and experience, we may find ourselves in disagreement and/or debate with one another. As such, it is important that we agree to conduct ourselves in a professional manner and that we work together to foster and preserve a virtual classroom environment in which we can respectfully discuss and deliberate controversial questions. I encourage you to confidently exercise your right to free speech—bearing in mind, of course, that you will be expected to craft and defend arguments that support your position. Keep in mind, that free speech has its limit and this course is NOT the space for hate speech, harassment, and derogatory language. I will make every reasonable attempt to create an atmosphere in which each student feels comfortable voicing their argument without fear of being personally attacked, mocked, demeaned, or devalued.

Any behavior (including harassment, sexual harassment, and racially and/or culturally derogatory language) that threatens this atmosphere will not be tolerated. Please alert me immediately if you feel threatened, dismissed, or silenced at any point during our semester together and/or if your engagement in discussion has been in some way hindered by the learning environment.

AI

The use of AI on projects is **not permitted**. All projects will be assessed against AI generated responses and cheating will be referred to the Office of Student Conduct and may result in major consequences such as a XF.

AI is allowed to be used for studying purposes only, there are benefits to using it to generate practice questions, explain concepts and give feedback. **Use of AI for studying specifically is encouraged**. If you want to discuss what is allowed further feel free to find me in office hours.

Piazza

This term we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TA, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza. If you have any problems or feedback for the developers, email team@piazza.com. You will be invited to the Piazza course using your university provided email by the end of the first week of classes at the latest, or by drop/add if you join the class late.

Major Assignments

Quizzes

- (Roughly) Weekly Check In Quizzes, 6-7 assignments total
 - **Online, on ELMS using Lockdown Browser**
 - **Meant to help you keep pace with the course, are due the Sunday course content is taught. (i.e. Quiz on Week 4 content will be due the Sunday of Week 4).**

Coding Projects

- Typically given every few weeks, roughly 6-7 projects
 - Due dates may vary, check ELMS and Piazza for specifics
- Implements coding/computing topics discussed in lecture and discussion
- Random spot checks will be given to ensure all students know the content.

Midpoint Oral Exam

- Oral Exams, with flexible scheduling, on Zoom
- You will need to sign up for a time with myself or a TA, all students must take the exam between weeks 5-7.
- Will mimic a coding interview that would be given to an entry level web developer.

Final Project

- Due by July 24th (Last Day of Classes)
- More details to come soon

Grading Structure

Assignment	Percentage %
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Programming Assignments	30%
Quizzes	20%
Midterm Exams	20%
Final Project	30%
Total	100%

Late Policy

Each student is provided with two 48-hour extensions for the programming assignments (i.e. not the quizzes, midterm exams, or final project). These extensions will be applied automatically to the first three late submissions (i.e. students do not choose which assignments use up their late allowance). No request (via email or otherwise) is necessary. After two late submissions, no late work will be accepted (i.e. any late work will be given a 0% and will not be graded). **You may only use one 48-hour extension at a time.**

Academic Integrity

Describe your expectations for academic integrity. The following is sample text regarding the use of Turnitin:

For this course, some of your assignments will be collected via Turnitin on our course ELMS page. I have chosen to use this tool because it can help you improve your scholarly writing and help me verify the integrity of student work. For information about Turnitin, how it works, and the feedback reports you may have access to, visit [Turnitin Originality Checker for Students](#)

The University's Code of Academic Integrity is designed to ensure that the principles of academic honesty and integrity are upheld. In accordance with this code, the University of Maryland does not tolerate academic dishonesty. Please ensure that you fully understand this code and its implications because all acts of academic dishonesty will be dealt with in accordance with the provisions of this code. All students are expected to adhere to this Code. It is your responsibility to read it and know what it says, so you can start your professional life on the right path. **As future professionals, your commitment to high ethical standards and honesty begins with your time at the University of Maryland.** It is important to note that course assistance websites, such as CourseHero, are not permitted sources, unless the instructor explicitly gives permission for you to use one of these sites. Material taken or copied from these sites can be deemed unauthorized material and a violation of academic integrity. These sites offer information that might not be accurate and that shortcut the learning process, particularly the critical thinking steps necessary for college-level assignments. Additionally, students may naturally choose to use online forums for course-wide discussions (e.g., Group lists or chats) to discuss concepts in the course. However, collaboration on graded assignments is strictly prohibited unless otherwise stated. Examples of prohibited collaboration include: asking classmates for answers on quizzes or exams, asking for access codes to clicker polls, etc. Please visit the [Office of Undergraduate Studies' full list of campus-wide policies](#) and reach out if you have questions.

Finally, on each exam or assignment you must write out and sign the following pledge:

"I pledge on my honor that I have not given or received any unauthorized assistance on this exam/assignment."

If you ever feel pressured to comply with someone else's academic integrity violation, please reach out to me straight away. Also, ***if you are ever unclear*** about acceptable levels of collaboration, ***please ask!*** To help you avoid unintentional violations, ***the following table*** lists levels of collaboration that are acceptable for each graded exercise.

	 OPEN NOTES	 SEARCH ONLINE	 ASK FRIENDS	 WORK IN GROUPS
In-class and in-lab ungraded coding tasks	✓	✓	✓	✓
Quizzes	✓	✗	✗	✗
Homeworks	✓	✓	✓	✓
Coding Projects	✓	✗	✓**	✗
Exams	✗	✗	✗	✗

****For coding projects you are encouraged to talk to others about high level aspects of your projects, no code or specific approaches should be shared. All work must be your own.**

Grades

All assessment scores will be posted on the course ELMS page. If you would like to review any of your grades (including the exams), or have questions about how something was scored, please email me to schedule a time for us to meet and discuss.

Late work will only be accepted in accordance with the late policy described above. I am happy to discuss any of your grades with you, and if I have made a mistake I will immediately correct it. Any formal grade disputes must be submitted in writing and within one week of receiving the grade.

Final letter grades are assigned based on the percentage of total assessment points earned. To be fair to everyone I have to establish clear standards and apply them consistently, so please understand that being close to a cutoff is not the same as making the cut (89.99 $\neq$ 90.00). It would be unethical to make exceptions for some and not others.

Final Grade Cutoffs							
+	97.00%	+	87.00%	+	77.00%	+	67.00%
		+				+	

A	93.00%	B	83.00%	C	73.00%	D	63.00%	F	<60.0%
-	90.00%	-	80.00%	-	70.00%	-	60.00%	-	

Course Topics

- HTML
- HTML APIs
- CSS
- Page Layouts (CSS Grid Layout, CSS Flexbox Layout)
- JavaScript
 - Variables, data types, expressions, operators
 - Conditional, iteration, statements
 - Functions, arrays
 - Prototypes, inheritance and Object-Oriented JS
 - DOM, events, events handling
 - Modules
 - JavaScript standard library
 - Iterators and generators
 - Asynchronous JavaScript
- Server-Side JavaScript
 - Introduction to Node.js
 - Express
 - Data persistence with MongoDB
- React
- JSX
- Miscellaneous

Resources & Accommodations

Accessibility and Disability Services

The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, on the basis of disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. The [Accessibility & Disability Service \(ADS\)](#) provides reasonable accommodations to qualified individuals to provide equal access to services, programs and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to make arrangements.

For assistance in obtaining an accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at adsfrontdesk@umd.edu. Information about [sharing your accommodations with instructors, note taking assistance](#) and more is available from the [Counseling Center](#).

Student Resources and Services

Taking personal responsibility for your own learning means acknowledging when your performance does not match your goals and doing something about it. I hope you will come talk to me so that I can help you find the right approach to success in this course, and I encourage you to visit [UMD's Student Academic Support Services website](#) to learn more about the wide range of campus resources available to you.

In particular, everyone can use some help sharpening their communication skills (and improving their grade) by visiting [UMD's Writing Center](#) and schedule an appointment with the campus Writing Center.

You should also know there are a wide range of resources to support you with whatever you might need ([UMD's Student Resources and Services website](#) may help). If you feel it would be helpful to have someone to talk to, visit [UMD's Counseling Center](#) or [one of the many other mental health resources on campus](#).

Basic Needs Security

If you have difficulty affording groceries or accessing sufficient food to eat every day, or lack a safe and stable place to live, please visit [UMD's Division of Student Affairs website](#) for information about resources the campus offers you and let me know if I can help in any way.

Veteran Resources

UMD provides some additional supports to our student veterans. You can access those resources at the office of [Veteran Student life](#) and the [Counseling Center](#). Veterans and active duty military personnel with special circumstances (e.g., upcoming deployments, drill requirements, disabilities) are welcome and encouraged to communicate these, in advance if possible, to the instructor.

Participation

- Given the interactive style of this class, attendance will be crucial to note-taking and thus your performance in this class. Attendance is particularly important also because class discussion will be a critical component for your learning.
- Each student is expected to make substantive contributions to the learning experience, and attendance is expected for every session.
- Students with a legitimate reason to miss a live session should communicate in advance with the instructor, except in the case of an emergency.
- Students who miss a live session are responsible for learning what they miss from that session.
- Additionally, students must complete all readings and assignments in a timely manner in order to fully participate in class.

Course Evaluation

Please submit a course evaluation through Student Feedback on Course Experiences in order to help faculty and administrators improve teaching and learning at Maryland. All information submitted to Course Experiences is confidential. Campus will notify you when Student Feedback on Course Experiences is open for you to complete your evaluations at the end of the semester. Please go directly to the [Student Feedback on Course Experiences](#) to complete your evaluations. By completing all of your evaluations each semester, you will have the privilege of accessing through Testudo the evaluation reports for the thousands of courses for which 70% or more students submitted their evaluations.

Copyright Notice

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